



UNIVERSAL BEAMS AND COLUMNS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	Ix		Iy		ix		iy		Zx		Zy	
in (mm)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W14 14 x 6 ³ / ₄ (356 x 171)	291	12,100	19.6	816	5.73	14.6	1.49	3.78	42.0	688	5.82	95.4
	340	14,200	23.3	970	5.83	14.8	1.53	3.89	48.6	796	6.91	113
	385	16,000	26.7	1,110	5.88	14.9	1.55	3.94	54.6	895	7.88	129
	469	19,522	32.7	1,362	5.95	15.1	1.57	3.99	65.5	1,073	9.60	157
W14 14 x 8 (356 x 203)	428	17,800	45.2	1,880	5.82	14.8	1.89	4.80	62.7	1,030	11.3	185
	485	20,200	51.4	2,140	5.85	14.9	1.91	4.85	70.3	1,150	12.8	210
	542	22,600	57.7	2,400	5.89	15.0	1.92	4.88	77.8	1,270	14.3	234
W14 14 x 10 (356 x 254)	640	26,600	107	4,450	5.98	15.2	2.45	6.22	92.2	1,510	21.5	352
	725	30,200	122	5,080	6.01	15.3	2.47	6.27	103	1,690	24.3	398
	796	33,100	134	5,580	6.04	15.3	2.48	6.30	112	1,840	26.6	436
	882	36,700	148	6,160	6.05	15.4	2.48	6.30	123	2,020	29.3	480
W14 14 x 14 ¹ / ₂ (356 x 368)	966	40,250	350	14,560	6.15	15.6	3.70	9.40	138	2,263	48.2	790
	999	41,600	362	15,100	6.14	15.6	3.70	9.40	143	2,340	49.9	818
	1,110	46,200	402	16,700	6.17	15.7	3.71	9.42	157	2,570	55.2	905
	1,166	48,530	420	17,470	6.21	15.8	3.72	9.45	164	2,681	57.6	944
	1,240	51,600	447	18,600	6.22	15.8	3.73	9.47	173	2,830	61.2	1,000
	1,373	57,160	492	20,470	6.26	15.9	3.75	9.52	189	3,104	67.1	1,099
	1,380	57,400	495	20,600	6.24	15.8	3.74	9.50	190	3,110	67.5	1,110
	1,530	63,700	548	22,800	6.28	16.0	3.76	9.55	209	3,420	74.5	1,220
	1,593	66,310	568	23,630	6.31	16.0	3.77	9.58	216	3,540	77.0	1,262
W14 14 x 16 (356 x 406)	1,710	71,200	677	28,200	6.33	16.1	3.98	10.1	232	3,800	87.3	1,430
	1,900	79,100	745	31,010	6.40	16.2	4.00	10.2	253	4,153	95.8	1,570
	1,900	79,100	748	31,100	6.38	16.2	4.00	10.2	254	4,160	96.2	1,580
	2,140	89,100	838	34,900	6.43	16.3	4.02	10.2	281	4,600	107	1,750
	2,400	99,900	931	38,800	6.50	16.5	4.05	10.3	310	5,080	119	1,950
	2,660	111,000	1,030	42,900	6.55	16.6	4.07	10.3	338	5,540	130	2,130
	2,942	122,500	1,124	46,820	6.62	16.8	4.10	10.4	368	6,028	141	2,324
	3,010	125,000	1,150	47,900	6.63	16.8	4.10	10.4	375	6,150	145	2,380
	3,400	142,000	1,290	53,700	6.71	17.0	4.13	10.5	415	6,800	161	2,640
	3,526	146,800	1,331	55,420	6.74	17.1	4.14	10.5	427	7,005	166	2,772
	3,840	160,000	1,440	59,900	6.79	17.2	4.17	10.6	459	7,520	179	2,930
	4,330	180,000	1,610	67,000	6.88	17.5	4.20	10.7	506	8,290	199	3,260
	4,399	183,100	1,631	67,910	6.90	17.5	4.20	10.7	512	8,390	201	3,294
	4,900	204,000	1,810	75,300	6.98	17.7	4.24	10.8	559	9,160	221	3,620
	5,440	226,000	1,990	82,800	7.07	18.0	4.27	10.8	607	9,950	241	3,950
	6,000	250,000	2,170	90,300	7.16	18.2	4.31	10.9	656	10,700	262	4,290
	6,600	275,000	2,360	98,200	7.26	18.4	4.34	11.0	706	11,600	283	4,640
	7,190	299,000	2,560	107,000	7.33	18.6	4.38	11.1	756	12,400	304	4,980
	8,210	342,000	2,880	120,000	7.48	19.0	4.43	11.3	838	13,700	339	5,560
	9,430	392,000	3,250	135,000	7.63	19.4	4.49	11.4	931	15,300	378	6,190
	10,800	450,000	3,680	153,000	7.80	19.8	4.55	11.6	1,040	17,000	423	6,930
W16 16 x 5 ¹ / ₂ (406 x 140)	301	12,500	9.59	399	6.26	15.9	1.12	2.84	38.4	629	3.49	57.2
	375	15,600	12.4	516	6.41	16.3	1.17	2.97	47.2	773	4.49	73.6